Experiment - 2

Aim: Convert the ER diagram into a relational schema. Write complete MySQL. Create table statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON Delete CASCADE and ~~not NULL~~ SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

Relational Schema:

Customer (
- customer_ID, PK,
- Name,
- Email UNIQUE,
- Phone_No,
- DOB

)

Address (
- Address ID Pk,
- CustomerID FK,
- Pincode,
- State,
- City

)

Seller (
- SellerID Pk,
- SellerName,
- PhoneNo Unique

)

Category (
- CategoryID Pk,
- CategoryName Unique,
- Description

)

Product (
ProductID Pk,
CategoryID Fk,
Price,
Image
)

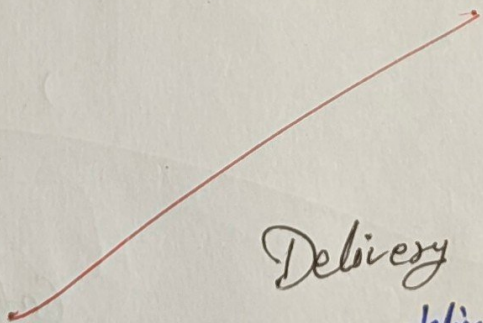
Electronic (
ProductID PK,
FK
)

Fashion (
productID PK
FK
)

Books (
productID PK,
FK,
SellerID FK.
)

Order (
orderID Pk,
CustomerID Fk,
orderDate,
OrderStatus
)

Order_item (
orderID PK/FK,
item_sq_no PK,
ProductID FK,
Quantity,
Unit_Price
)



Payment (
paymentID Pk,
orderID FK Unique,
PaymentDate,
Amount,
PaymentMethod,
status
)

Delivery (
deliveryID Pk,
orderID FK Unique,
CourierName,
Status,
DeliveryDate
)

Source Code :

```
Create DATABASE EcommerceDB;
Use EcommerceDB;

CREATE TABLE Customer(
        CustomerID INT Primary key Auto_increment,
        Name varchar(100) NOT NULL,
        Email varchar(100) NOT NULL Unique,
        PhoneNo varchar(15) NOT NULL,
        DOB DATE NOT NULL
);

CREATE TABLE Address(
        AddressID INT Primary key Auto_increament,
        CustomeID Int Not null,
        Pincode varchar(10) NOT NULL,
        State varchar(50) not null,
        City varchar(50) not null,
        Foreign key (customerID) reference Customer
        (customerID) on Delete CASCADE
);

Create Table Category(
        categoryID int Primary key Auto_increament,
        CategoryName varchar(100) not null unique,
        Description Varchar(250)
);
```

```
Create Table Seller (
        SellerID int Primary key auto_increament,
        SellerName varchar(100) Not null,
        PhoneNo varchar(20) not null unique
);

Create Table Product (
        ProductID int Primary key auto_increament,
        CategoryID int Not null,
        Price Decimal (10,2) not null,
        Image varchar (255),
        Foreign key (CategoryID) refrences category
        (CategoryID) on Delete CASCABE .
);

Create Table orders (
        orderID int Primary key auto_increament,
        customerID int Not null,
        orderDate Date Not null,
        orderStatus varchar(30) not null,
        Foreign key (customerID) refrences customer
        (customerID) on Delete CASCABE
);
```

```
Create Table OrderItem (
        orderID int not null,
        item_seq_no int not null,
        ProductID int not null,
        Quantity int not null,
        Unit_Price Decimal (10,2) not null,
        Primary key (OrderID, item_seq_no),
        foreign key (OrderID) reference order (orderID)
            on DELETE CASEADE
        foreign key (ProductID) reference Product (ProductID)
            ON Delete CASCADE
);
```

```
Create Table Payment (
        PaymentID int Primary key auto_increament,
        OrderID int Not null unique,
        Payment_Date Date Not Null,
        Amount Decimal (10,2) Not Null,
        Status varchar (30) Not null,
        Foreign key (orderID) reference order (orderID)
            on Delete CASCADE
);
```

```
Create Table Delivery (
    DeliveryID int Primary key auto_increamment,
    orderID int not null unique,
    CourierName varchar(100) not null,
    Status varchar(30) not null,
    DeliveryDate Date,
    Foreign key (orderID) references orders(orderID)
        on Delete CASCADE.
);
```

Inserting Sample Data —

```
Insert into Customer (Name, Email, PhoneNo, DOB) values
("Sachin", "Sachin@gmail.com", "9395972010", "2025-05-15"),
("Sahil", "Sahil@gmail.com", "9839105025", "2004-08-20");

Insert into Category (CategoryName, Description) values
("Electronics", "Electronic Products"),
("Fashion", "Fashion Product"), ("Books", "Books")

Insert into seller (SellerName, PhoneNo) values
("Tech Store", "90000001"),
("Book Store", "90000002");
```

```
insert into Product (categoryID, Price, Image) values
(1, 5000, "Laptop.jpg"), (2, 999, "Shirt.png"),
(3, 599, "Book.jpg");

insert into Electronic values (1);
insert into Fashion values (2);
insert into Books values (3,2);

Insert into Address (customerID, PinCode, State, City) values
(1, "200001", "UP", "Kanpur"),
(2, "110001", "Delhi", "Delhi");

Insert into orders (customerID, orderDate, OrderStatus) values
(1, "2026-08-19", "Placed"),
(2, "2026-08-19", "Shipped");

Insert into OrderItem values
(1,1,1,1,5000), (2,1,2,2,999);

Insert into Delivery (orderID, CourierName, Status,
                      DeliveryDate) values
(1, "Delivery", "Delivered", "2026-08-20"),
(2, "DTDC", "Pending", Null);
```

Referential integrity Violations —

```
insert into orders (customerID, OrderDate, Orderstatus)
values
(999, "2026-08-19", "Placed");
```

Output :-

```
ERROR: Cannot add or update a child row:
    a. foreign key constraint fails
```

Unique Constraint Violation —

```
Insert into ~~cust~~ customer (Name, Email, PhoneNo, DOB)
Values
("Test", "Sachin@gmail.com", "9898989898", "2025-01-01");
```

Output :-

```
Error: Duplicate entry "[redacted]Sachin@gmail.com"
    for key 'customer.Email'
```

```
Select * from Customer;
Select * from Product;
Select * from Orders;
Select * from OrderItem;
```

ON DELETE SET NULL

```
Delete from Seller where sellerID = 2;

Select * from Books;
```

ON DELETE CASCADE

```
Delete from Customer where CustomerID = 2;
Select * from Order where CustomerID = 2;
```

Output :-

Customer ID	Name	Email	Phone No	DOB
1	Sachin	Sachin@gmail.com	9897324592	2005-05-20
2	Sahil	Sahil@gmail.com	9839105025	2005-05-15

ProductID	CategoryID	Price	image
1	1	5000.00	Laptop.png
2	2	999.00	Shirt.png
3	3	599.00	book.png

OrderID	CustomeID	OrderDate	orderStatus
1	1	2026-08-19	Placed
2	2	2026-08-19	Shipped

OrderID	item_seq_no	ProductID	quantity	Unit_price
1	1	1	1	50000.00
2	1	2	2	999.00

ProductID	SellerID
3	Null

Empty set